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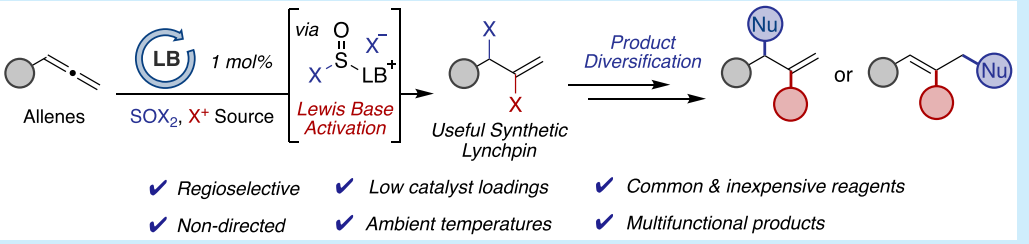
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Lewis Base-Catalyzed Interhalogenation of Terminal Allenes: Selective Strategy for Accessing Vicinal Vinylic, Allylic Heterodihalides

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ABSTRACT: Though precedent remains limited, the selective interhalogenation of allenes offers a valuable synthetic strategy to access products where each halide exhibits orthogonal reactivity. Here, we describe a Lewis base-catalyzed approach for the dihalogenation (bromochlorination, iodochlorination, iodobromination, and dibromination) of terminal allenes. By employing just 1 mol % of triphenylphosphine oxide or hexamethylphosphoramide to activate thionyl halides in the presence of electrophilic halogenation reagents, we achieve the conversion of monosubstituted allenes to vicinal allylic, vinylic dihalides with up to 93% yield and >20:1 regioisomeric ratio, favoring the branched dihalogenated product. A range of functional groups is tolerated, including nitrile, ester, phosphate, sulfonamide, and silyl groups, and the reaction proved to be scalable. The utility of various dihalide products was investigated in substitution and cross-coupling chemistry, highlighting the distinct reactivity among the different classes.

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The utility of organohalides is evident from their widespread use in synthetic chemistry, pharmaceuticals, and agrochemicals.[1−4] With the continual discovery of bioactive, organohalogen natural products, many of which feature vicinal heterodihalide motifs, designing elegant approaches for the incorporation of halogens into organic frameworks is of increasing importance. Consequently, much progress has been made over the past decade, including recent work by our group, for the preparation of heterodihalides from alkenes and alkynes (Figure 1a).[5,6]

By comparison, the interhalogenation of allenes is hardly developed, despite recent advances in their selective functionalization (Figure 1b).[7] Further, the utility of the resulting heterodihalide products is apparent when one considers the divergent reactivity profiles innate to allylic and vinylic halides (e.g., nucleophilic displacement, cross coupling chemistry, etc.) and the broad utility of alkenes as synthetic handles. For these reasons, vicinal vinylic, allylic heterodihalides, though underexplored, bear the capacity to serve as prized synthetic lynchpins.

Strategies for allene interhalogenation are generally plagued by issues of regio-, stereo-, and site-selectivity.[7e,8] The onerous nature of these transformations is exemplified in the limited number of reports detailing the bromochlorination of allenes (Figure 1b). Two independent examples disclosed the use of BrCl for the bromochlorination of propadiene or 2,3-pentadiene.[9] However, these reactions suffer from low yields and the presence of solvent-incorporated byproducts. In some cases, solvolysis represented the major reaction pathway or no yields were reported altogether, raising concerns as to the overall efficiency of this transformation. During the course of our studies, Katayev and co-workers disclosed a unique transfer dihalogenation strategy applicable to allenes. However, only a singular example of vicinal bromochlorination on an aryl-substituted allene was demonstrated.[10]

We sought to develop a strategy for the interhalogenation of terminal allenes by utilizing sulfinyl halides, thereby permitting catalyst control in the availability of the nucleophilic halide anion (Figure 1c). Based on our prior work and the studies of others, we surmised that a transient methylenehaliranium species (**II**) could be intercepted by a halide anion salt (**III**) generated through Lewis base activation of thionyl chloride or thionyl bromide.[6,11,12] This strategy would allow for the

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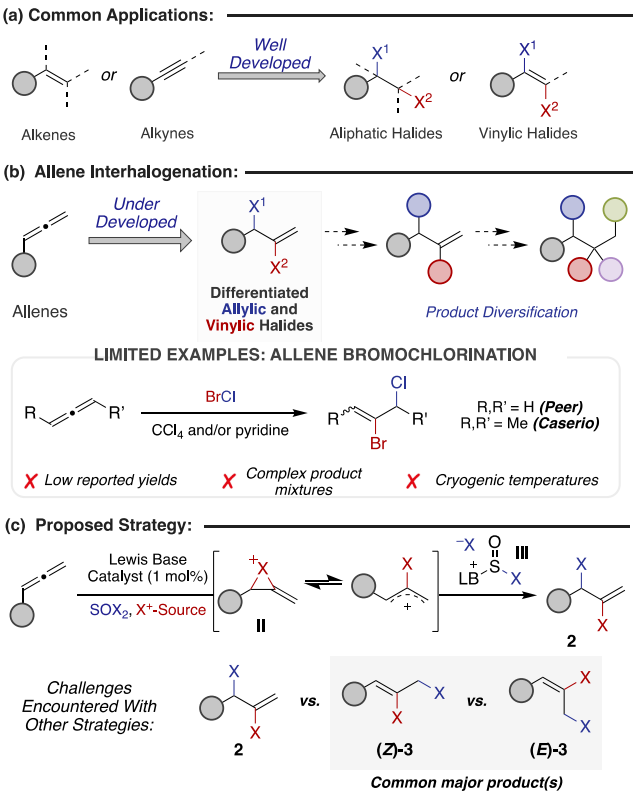


Figure 1. Allene interhalogenation—prevalence, strategy, and limitations.

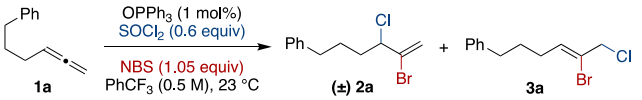
precise installation of two different halides across the three consecutive sp^2-hybridized carbons of an allene to furnish dihalogenated product **2**. To date, common methods employed for allene dihalogenation are either unselective or primarily feature linear product (**3**) selectivity with varying levels of stereoselectivity.[13,14]

We commenced our investigations by surveying conditions for allene bromochlorination. When 4,5-hexadienylbenzene (**1a**) was subjected to thionyl chloride and a slight excess of *N*-bromosuccinimide (NBS) in the absence of a catalyst, only trace quantities of isomeric bromochlorides **2a** and **3a** were observed (Table 1, entry 1). Instead, a significant amount of undesired dibromination byproduct was observed. Under the same conditions, but in the presence of 1 mol % triphenylphosphine oxide, product was formed in 82% yield, favoring branched bromochloride **2a** with a 13:1 regioisomeric ratio (rr) (entry 2).

While various solvents were generally suitable, trifluorotoluene proved optimal with respect to reaction efficiency and selectivity (entries 2−4).[15] A survey of Lewis base catalysts revealed triphenylphosphine oxide and hexamethylphosphoramide (HMPA) as superior catalysts, delivering up to 85% yield (18:1 rr) at minimal loadings (entries 5−8). In the examination of alternative bromenium sources, *N*-bromophthalimide (NBP) provided optimal results, particularly when used in combination with HMPA, affording bromochloride **2a** in 93% yield with 17.5:1 rr (entries 9−11). Further, the reaction displays similar efficiency under noninert conditions (entry 12).

With the optimized conditions established, we next examined the scope of this transformation (Table 2). In all cases, compound **2** was obtained as the major product, with

Table 1. Reverse Optimization of Allene Bromochlorination[a]



Entry	Variation from std. conditions	Conv. (%)	Yield (%)[b]	2a:3a
1	No $OPPh_3$	75	6	-
2	**None**	**>99**	**82**	**13:1**
3	Using PhMe as solvent	73	50	12.5:1
4	Using CH_2Cl_2 as solvent	>99	73	12.5:1
5	PPh_3 instead of $OPPh_3$	86	73	11:1
6	DMAP instead of $OPPh_3$	92	56	15:1
7	HMPA instead of $OPPh_3$	98	85	18:1
8	TPPA instead of $OPPh_3$	91	71	17:1
9	NBP instead of NBS	97	88	14:1
10[c]	DBDMH instead of NBS	95	58	12:1
11	**HMPA and NBP**	**>99**	**93[d]**	**17.5:1**
12	Non-dried glassware, under air	>99	83	17:1

[a]Reactions were carried out under an atmosphere of N_2 using 0.2−0.3 mmol of **1a** (1.0 equiv). Reactions were run for a duration of 3−4 h. DMAP = 4-dimethylaminopyridine, HMPA = hexamethylphosphoramide, TPPA = tripyrrolidinophosphoric acid triamide, NBP = *N*-bromophthalimide, DBDMH = 1,3-dibromo-5,5-dimethylhydantoin. [b]Yields and regiomeric ratios were determined by ^{1}H NMR spectroscopy of the crude reaction mixture, using mesitylene (0.1 mmol) as an internal standard. [c]0.52 equiv of DBDMH was used instead. [d]Represents the isolated yield on a 1 mmol scale.

varying levels of linear-to-branched regioselectivity. Monosubstituted allenes, both with and without alpha-branching substituents, provided similar regioisomeric ratios (rr), yielding branched dihalide products (**2a**, **2b**, and **2d**). Furthermore, the reaction demonstrated scalability, allowing for the isolation of 2.9 g of **2a** with minimal loss of efficiency and selectivity. One exception, aryl allene **1c**, resulted in both poor yield and selectivity, likely due to conjugation modulating the reactivity of the internal alkene. However, this strategy enabled the synthesis of the branched bromochloride product **2c** as the major isomer, notably in contrast to previous methods that typically favor the formation of linear dihalide products.

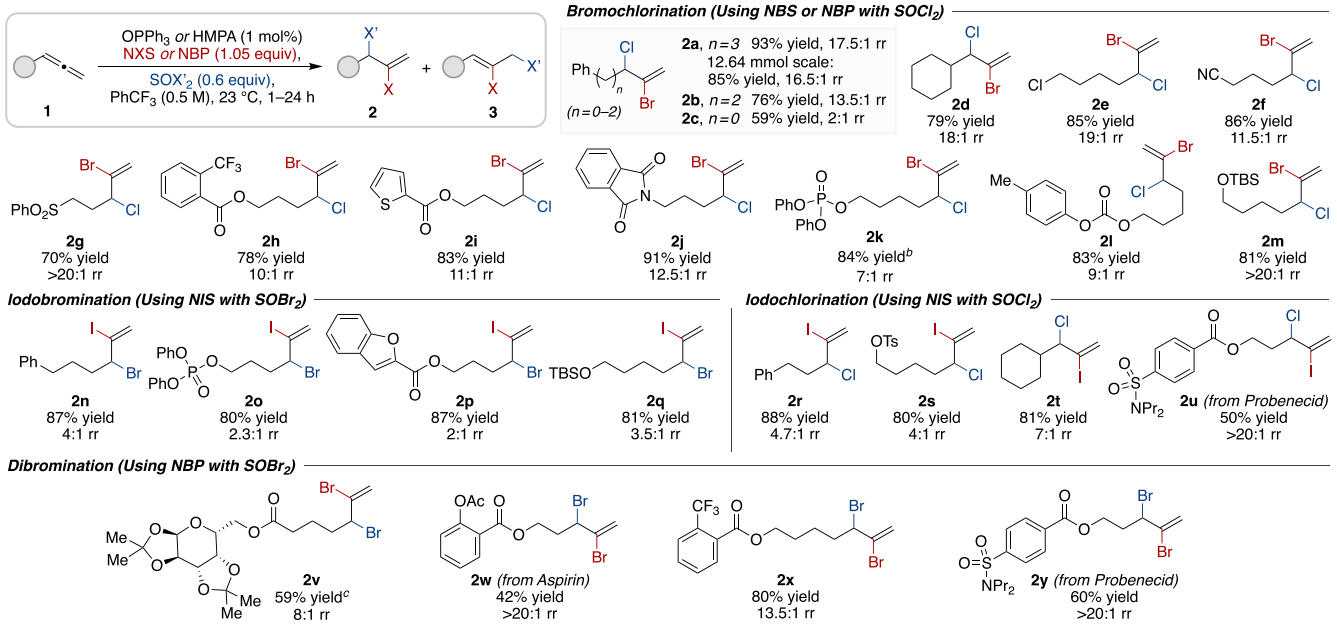
Allenes bearing a variety of functional groups were well-tolerated regardless of their proximity to the reactive site (**2e**−**2m**).[16] In all cases, the desired products were obtained in good-to-high yields with generally high selectivity. Though many products were isolated with improved yield and/or selectivity using the HMPA/NBP conditions (**2a**, **2b**, **2j**, and **2m**), some substrates (e.g., **1h** and **1i**) performed better with the $OPPh_3$/NBS conditions instead. Furthermore, substrate **1k**, which contains a Lewis basic phosphate substituent, did not require the addition of a catalyst.

By modifying the electrophilic and nucleophilic halogenating reagents to *N*-iodosuccinimide (NIS) and $SOBr_2$ *or* NIS and $SOCl_2$, a range of iodobrominated and iodochlorinated products were isolated (**2n**−**2u**), albeit with reduced selectivity. For comparison, optimization substrate **1a** yielded iodobromide **2n** in 87% yield with a 4:1 rr and iodochloride **2r** in 88% yield with a 4.7:1 rr. Notably, although allene **1u**, derived from the uric acid reducer probenecid, formed a crude product mixture with a 1.5:1 ratio of **2u** to **3u**, the branched product could be cleanly isolated, yielding iodochloride **2u** as a pure isomer in 50% yield.

Historically, methods for the dibromination of allenes have been plagued by poor yields and/or selectivity.[13b,17] Therefore,

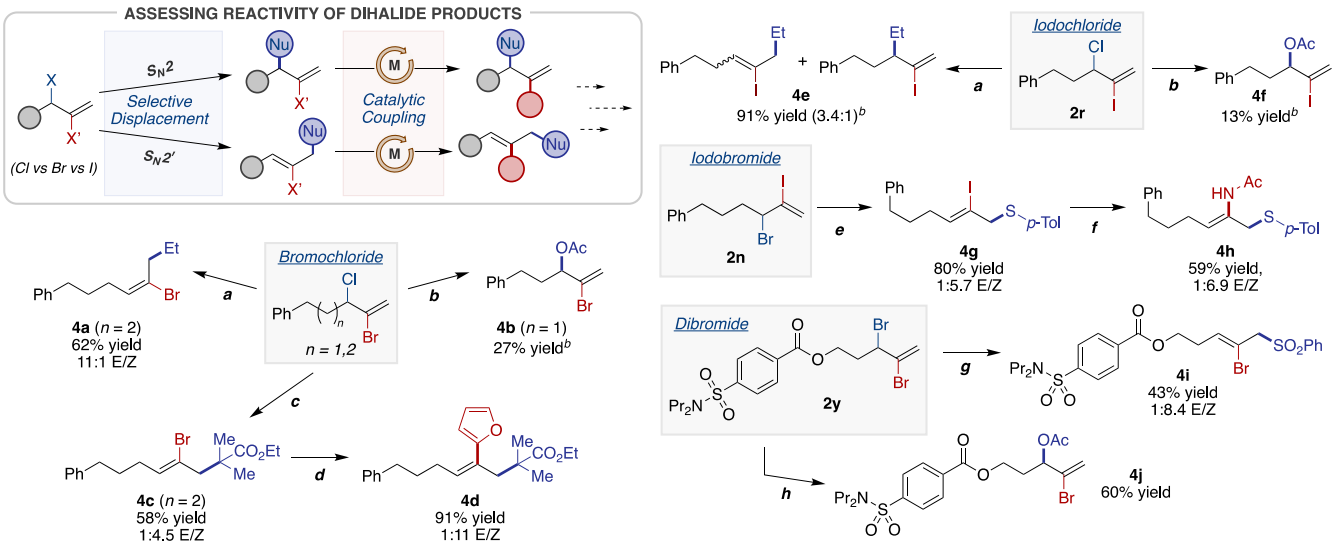
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Table 2. Scope of Catalytic Allene Dihalogenation[a]



[a]Unless specified otherwise, reactions were carried out under a N_2 atmosphere on a 0.5−12.64 mmol scale (see the Supporting Information). Regiomeric ratios (rr) were determined by analysis of the ^{1}H NMR spectra of the purified products. Reported yields represent the combined mixture of regioisomers. [b]Reaction was carried out without addition of catalyst. [c]Reaction was carried out using NBP as the limiting reagent instead.

Scheme 1. Diversification of Interhalides[a]



[a]Unless specified otherwise, reactions were carried out under a N_2 atmosphere. Regiomeric ratios (rr) were determined by analysis of the ^{1}H NMR spectra of the purified products. E/Z selectivity was determined via NOESY analysis. [b]Yield was determined by ^{1}H NMR spectroscopy of the crude reaction mixture, using mesitylene (0.1 mmol) as an internal standard. **Conditions a:** CuCN (5 mol %), EtMgBr (1.2 equiv), CH_2Cl_2 (0.1 M), −78 °C, 22−23 h. **Conditions b:** AcOK (7.0 equiv), 18-crown-6 (7.5 equiv), DMF (0.025 M), 23 °C, 21 h. **Conditions c:** *n*-BuLi (1.1 equiv), $HNEt_2$ (1.2 equiv), ethyl isobutyrate (1.0 equiv), THF (0.5 M), −78 to 23 °C, 22 h. **Conditions d:** $Pd(PPh_3)_4$ (5 mol %), 2-(tributylstannyl)furan (1.2 equiv), 1,4-dioxane (0.2 M), 100 °C, 15 h. **Conditions e:** NEt_3 (1.0 equiv), *p*-thiocresol (1.0 equiv), benzene (0.25 M), 12 h. **Conditions f:** CuI (5 mol %), Cs_2CO_3 (1.5 equiv), acetamide (1.2 equiv), *N,N*-dimethylethylenediamine (10 mol %), THF (0.33 M), 70 °C, 17 h. **Conditions g:** $PhSO_2Na$ (4.0 equiv), $Pd(PPh_3)_4$ (5 mol %), DMSO (375 μL), THF (0.2 M), 60 °C, 17 h. **Conditions h:** AcOK (2.0 equiv), 18-crown-6 (2.5 equiv), DMF (0.3 M), 23 °C, 1 h.

we were interested in evaluating the selectivity of this catalytic manifold by comparison. Using NBP with thionyl bromide under the standard catalytic conditions developed here, the dibromination of allenes **1v**−**1y** largely favored halogenation at the internal alkene, including substrates featuring substructures derived from marketed therapeutics (**1w**, **1y**). These results demonstrate, in some cases, the orthogonal selectivity imparted by this catalyst-controlled strategy (see Supporting Information for additional examples).

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To investigate the utility of these uncommon vicinal vinylic, allylic heterodihalides in organic synthesis, we chose to explore the innate reactivity differences between each class of dihalide (Scheme 1).[18] Focusing initially on bromochlorides **2a** and **2b**, we applied copper-catalyzed S_N2' allylic alkylation, yielding **4a** in 62% with an E/Z ratio >10:1.[19] In contrast, S_N2 displacement proved more challenging, though acetate **4b** was obtained in low yield.[20] To illustrate the utility of these products for consecutive functionalization, enolate alkylation to afford vinyl bromide **4c** followed by Stille cross-coupling gave an overall two-step yield of 53% of ester **4d**.

Comparatively, the analogous iodochloride **2r** demonstrated poorer selectivity both in the copper-catalyzed allylic alkylation (**4e**) and the direct substitution with KOAc (**4f**). A limitation of the iodohalides, in general, is their bench instability, necessitating rapid purification of the dihalide followed by immediate setup for subsequent diversification reactions. Nevertheless, iodobromide **2n** exhibited superior S_N2' reactivity with *p*-thiocresol (**4g**, 80% yield 1:5.7 E/Z) followed by copper-catalyzed amide coupling of the vinyl iodide to afford enamide **4h** in 59% isolated yield.[21,22]

As anticipated, vicinal vinylic, allylic dibromides proved to be the best compromise between the bromochlorides and the iodohalides, owing to the lability of the allylic bromide combined with the better stability of the vinyl bromide. Probenecid-derived dibromide **2y** could undergo highly selective regiodivergent substitutions to afford vinyl bromides **4i** and **4j** in synthetically useful yields. Notably, palladium-catalyzed Tsuji-Trost substitution with benzenesulfinate was feasible without significant *β*-halo elimination to the allene, as observed with other dihalide variants.[23]

In conclusion, we have developed a catalytic dihalogenation of monosubstituted allenes, leading to vicinal vinylic and allylic dihalides. Products were isolated in up to 93% yield and >20:1 regioselectivity using common electrophilic reagents in combination with thionyl halides as latent nucleophiles activated by simple Lewis base catalysts. This convenient strategy for the synthesis of bromochlorides, iodochlorides, iodobromides, and dibromides employs noncryogenic reaction temperatures and low catalyst loadings of 1 mol %. The resulting dihalides were evaluated as useful building blocks for the preparation of a diverse range of functionalized di- and trisubstituted alkenes.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00422.

General procedural information, mechanistic studies, characterization data and spectra (PDF)

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Notes

The authors declare no competing financial interest.

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